

Register No.:

1546
October 2025

Time – Two hours
(Maximum Marks: 60)

- N.B.** 1. Answer all questions under Part A. Each question carries 1 mark.
2. Answer any 5 questions under Part B. Each question carries 2 marks.
3. Answer any 3 questions under Part C. Each question carries 10 marks.

PART – A

1. System administrators are responsible for monitoring:
 - a) Social media posts
 - B) System performance and uptime
 - C) Company profits
 - D) Product advertisements

2. What does ethics in system administration primarily involve?
 - a) Advertising company products
 - b) Maintaining professional appearance
 - c) Using admin privileges responsibly and protecting data privacy
 - d) Increasing system speed at all costs

3. What is one key feature introduced in Windows 10?
 - a) Removal of the Start Menu
 - b) Internet Explorer as the default browser
 - c) Virtual desktops and Cortana
 - d) Lack of update support

4. What is the first step in the installation procedure of an operating system?

- a) Formatting a USB drive
- b) Connecting to Wi-Fi
- c) Booting from installation media
- d) Installing antivirus software

5. Which file extension is commonly used for installing software in windows.

- a) .exe
- b) .apk
- c) .dmg
- d) .bat

6. What is the Event Viewer used for in Windows?

- a) To install new software
- b) To view system errors and warnings
- c) To create backups
- d) To update the operating system

7. Which Windows tool shows detailed system performance metrics like CPU, memory, disk, and network usage?

- a) Resource Monitor
- b) Command Prompt
- c) Quick Assist
- d) Run

8. An application on your PC is freezing frequently. What tool can you use to check which processes are using the most CPU or memory?

- a) Resource Monitor or Task Manager
- b) Event Viewer
- c) Disk Cleanup
- d) Windows Backup utility

9. Which command displays the contents of a file?

- a) view
- b) see
- c) cat
- d) open

10. Which symbol represents the current directory in Linux?

- a) /
- b) ~
- c) .
- d) . .

11. The top-level directory in a Linux file system is called the:

- a) /home
- b) /root
- c) /bin
- d) /** (slash)**

12. The /etc directory mainly stores:

- a) User files
- b) System configuration files
- c) Temporary files
- d) Log files

13. What does it indicate if the command ping 8.8.8.8 works but ping www.google.com fails?

- a) The network interface is disabled
- b) The DNS service is not functioning properly
- c) The system is offline
- d) The router is not connected

14. The command `free -m` is used to display:

- a) Disk usage in megabytes
- b) Memory usage in megabytes
- c) CPU utilization percentage
- d) Network usage in MB

15. A shell script is usually written in which language?

- a) Python
- b) Bash
- c) Java
- d) C++

16. In a script, you want to stop execution immediately when any command fails. Which setting should you apply?

- a) `set -e`
- b) `set +x`
- c) `trap exit`
- d) `exit 0`

17. When a DHCP client sends a DHCP Discover message, it is attempting to:

- a) Confirm an assigned IP address
- b) Request a lease renewal
- c) Locate a DHCP server in the network
- d) Release its IP address

18. When a device sends an ARP request, who responds?

- a) The router only
- b) The DHCP server
- c) The device with the matching IP address
- d) All devices in the network

19. Which topology provides the best fault tolerance?

- a) Bus
- b) Ring
- c) Mesh
- d) Star

20) A signature-based IDS detects attacks by:

- a) Learning from past traffic behavior
- b) Comparing traffic to known attack patterns
- c) Generating random detection rules
- d) Blocking unrecognized traffic automatically

PART – B

21) Mention three reasons why system administration is important.

22) Differentiate between Windows Backup and System Restore.

23) How will you troubleshoot errors caused by a newly installed application?

24) Compare /home and /root directories in Linux.

25) Apply a command to list all installed packages on a YUM-based system.

26) Why is shell scripting important for automating Linux tasks?

27) How does analysing script output help in debugging?

28) What is the working process of DHCP?

PART – C

29) Explain the different types of operating systems

30) Explain the installation and configuration of Antivirus software.

31) Analyse the relationship between user, group, and permission management in Linux security administration.

- 32) Examine how system resource monitoring tools help in diagnosing performance bottlenecks.
- 33) Compare and contrast the characteristics of TCP and UDP protocols.
